

EMIOS – Enterprise Migration Intelligence Operating System

Vision

Instead of asking *How many hours will migration take?*, we're answering *How should this enterprise evolve safely and profitably?*

Problem Statement

The Core Gap: Enterprise migration assessments are structurally flawed because teams completely lack visibility into the legacy ecosystem.

Primary Failure Drivers: Traditional manual discovery methods fail to capture critical environment variables, commonly leading to:

- Hidden dependencies
- Undocumented business rules
- Tribal knowledge
- Unknown downstream impacts
- Poor migration sequencing
- Incomplete risk visibility

The Downstream Ripple Effect: This severe information gap directly slows down cloud adoption velocity and dramatically increases migration failure rates, encouraging a costly culture of inaction.

The Pre-Sales Resource Bottleneck: Traditional pre-sales assessments lock down a scarce team of 3–5 senior architects for 4–6 weeks per enterprise client. This demands a massive, high-risk upfront engineering investment long before any migration execution begins.

The Market Requirement: Organizations desperately need an intelligent platform that maps enterprise landscapes, simulates migration outcomes, predicts risks, and recommends data-driven modernization strategies before migration begins.

What Makes EMIOS Different

Existing Tools	EMIOS
Static dependency scanners	Living Digital Twin that evolves as the system changes
Siloed risk reports	Collaborative multi-agent system where agents negotiate trade-offs
Consultant-driven estimates	Simulation-first planning with Monte Carlo confidence intervals
Black-box recommendations	Explainable AI that shows <i>why</i> a path was chosen
One-time assessments	Self-learning knowledge base that preserves organizational memory

Proposed Solution

EMIOS creates a Digital Twin of enterprise systems and uses AI agents to autonomously discover, analyze, simulate, and plan migration initiatives.

Core Modules (Full Platform)

Module	Purpose
Enterprise Digital Twin	Real-time mirror of systems, data flows, and business logic
Migration Intelligence Graph	Knowledge graph connecting applications, dependencies, business rules, and risks
Autonomous Discovery Agent	Crawls codebases, configs, and documentation to map the landscape without manual input
Multi-Agent Planning Engine	Orchestrates specialized agents that negotiate migration sequencing and resource allocation
Simulation Engine	Runs “What-If” scenarios against the Digital Twin before any real changes occur
Monte Carlo Risk Prediction	Generates probabilistic risk distributions with confidence intervals for timeline and cost
Business Impact Engine	Quantifies revenue, compliance, and operational impact of each migration path
Explainable Recommendation Engine	Translates agent decisions into human-readable rationale with visual evidence
Self-Learning Knowledge Base	Captures tribal knowledge and improves recommendations with every migration
Executive Decision Copilot	Natural language interface for C-level stakeholders to explore scenarios

Recommended MVP

Pre-built before hackathon: Digital Twin visualization framework, Neo4j schema, and React dashboard shell.

Built During Hackathon

Component	Agentic Behavior
Migration Intelligence Graph	Auto-populated from uploaded metadata
Discovery Agent	Autonomously identifies undocumented services by analyzing connection patterns; flags “orphan” components for human review
Dependency Agent	Detects circular dependencies and proposes decoupling strategies; escalates critical paths to Planner Agent
Risk Agent	Independently calculates failure probability per component; negotiates timeline adjustments with Planner Agent when risk exceeds 15% threshold
Planner Agent	Generates migration waves by balancing risk, dependency order, and business priority; accepts or rejects Risk Agent proposals
What-If Simulation Engine	Executes migration scenarios against the Digital Twin; surfaces cascading failures
Explainable Recommendation Dashboard	Shows <i>why</i> each recommendation was made citing specific dependencies, risk scores, and agent negotiation logs

Success Criteria

1. Ingestion Capability: Successfully ingest raw migration metadata via CSV or API dump without manual data cleansing.
2. Digital Twin Generation: Dynamically render a functional enterprise system topology map within seconds of ingestion.
3. Agentic Discovery: Autonomously identify hidden architectural dependencies and circular loops via LangGraph multi-agent orchestration.
4. Risk Simulation: Execute "What-If" scenarios that visually calculate and propagate downstream failure risks across the architecture grid.
5. Value Quantification: Generate real-time financial savings projections and operational business impact metrics based on simulation data.
6. Explainable Roadmap: Output a multi-wave migration schedule where every decision is backed by transparent, human-readable agent rationale.

Tech Stack

Layer	Technology
Frontend	React + Material UI
Backend API	FastAPI
Agent Orchestration	LangGraph
LLM	Azure OpenAI / GPT-4o
Graph Database	Neo4j
Vector Store	Qdrant
Observability	Langfuse
Deployment	Docker + Azure

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